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CSE400 – Fundamentals of Probability in Computing

Lecture 10: Randomized Min-Cut Algorithm

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Outline

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Min-Cut Problem

Why use min-cut?

We use the min-cut algorithm in various applications to solve problems related to network connectivity, reliability, and optimization.

Network Design:
Min cut helps in improving the efficiency of communication and optimizing network flow. The algorithm is used in network design to find the minimum capacity cut.

Communication Networks:
For understanding the vulnerability of the networks to failures minimum cut can be useful. It helps building robust and fault-tolerant communication networks.

VLSI Design:
In Very Large Scale Integration (VLSI) design, the algorithm is useful for partitioning circuits into smaller components leading to reduced interconnectivity complexity.

Refer to Section 1.5. *Application: A Randomized Min-Cut Algorithm* in *Probability and Computing (2nd Edition).*

Min-Cut Problem

What is min-cut?

A cut-set in a graph is a set of edges whose removal breaks the graph into two or more connected components.

Given a graph $G = (V, E)$ with n vertices, the minimum cut – or min-cut – problem is to find a minimum cardinality cut-set in G.

Min-cut algorithms, such as Karger's algorithm, are random and can be sensitive to the initial choice of edges. If the algorithm happens to contract critical edges early in the process, it might find a smaller cut.

The main operation in the algorithm is edge contraction, which is operation that removes an edge from a graph while simultaneously merging the two vertices.

In contracting an edge (u, v) we merge the two vertices u and v into one vertex, eliminate all edges connecting u and v, and retain all other edges in the graph. The new graph may have parallel edges but no self-loops.

Min-Cut Problem

Successful Min-Cut Run

A successful min cut run refers to the success in the outcome of an algorithm designed to find the minimum cut in a graph.

(Figure: Successful Run)

Min-Cut Problem

Unsuccessful Min-Cut Run

An unsuccessful min-cut run refers to an iteration of a min-cut algorithm where the algorithm fails to correctly identify the minimum cut of a given graph.

(Figure: Unsuccessful Run)

Min-Cut Problem

Max-Flow Min-Cut Theorem

The max-flow min-cut theorem states that:
"In a flow network, the maximum amount of flow passing from the source to the sink is equal to the total weight of the edges in a minimum cut".

Capacity of a cut = the sum of the capacity of the edges in the cut that are oriented from a vertex $\in X$ to a vertex $\in Y$

Minimum cut = the cut in the network that has the smallest possible capacity.

Minimum cut capacity = the capacity of the minimum cut.

Maximum flow = the largest possible flow from source S to sink T.

Deterministic Min-Cut Algorithm

Stoer–Wagner Min-Cut Algorithm

Let s and t be two vertices of a graph G. Let $G/\{s, t\}$ be the graph obtained by merging s and t. Then a minimum cut of G can be obtained by taking the smaller of a minimum s-t-cut of G and a minimum cut of $G/\{s, t\}$.

The theorem holds since either there is a minimum cut of G that separates s and t, then a minimum s-t-cut of G is a minimum cut of G;
or there is none, then a minimum cut of $G/\{s, t\}$ does the job.

Pseudocode

Algorithm 1: MinimumCutPhase(G, a)

$A \leftarrow \{a\}$;
while $A \neq V$ do
  add to A the most tightly connected vertex;
return the cut weight as the “cut of the phase”;

Algorithm 2: MinimumCut(G)

while $|V| \geqslant 1$ do
  choose any a from V;
  MinimumCutPhase(G, a);
  if the cut-of-the-phase is lighter than the current minimum cut then
    store the cut-of-the-phase as the current minimum cut;
  shrink G by merging the two vertices added last;

return the minimum cut;

Randomized Min-Cut Algorithm

Why Randomized Algorithm?

Randomized algorithms provide a probabilistic guarantee of success. It provides a more accurate estimate of the minimum cut with fewer iterations.

Efficiency
Parallelization
Approximation Guarantees
Avoidance of Worst-Case Instances
Heuristic Nature
Robustness

Randomized Min-Cut Algorithm

Karger’s Randomized Algorithm

(Randomized Min-Cut Algorithm: Karger’s Randomized Algorithm)

Randomized Min-Cut Algorithm

Comparison: Deterministic Min-Cut vs Randomized Min-Cut Algorithm

Any specific problem is responsible for deciding the approach.

Deterministic Min Cut:
It always guarantees an exact minimum cut.
It may have higher time complexity for large graphs.
The Stoer-Wagner algorithm, a deterministic min cut algorithm, has a time complexity of
$O(V \cdot E + V^2 \log V)$, where V is the number of vertices and E is the number of edges.

Randomized Min Cut:
It provides an approximate minimum cut with high probability.
The Karger's algorithm, a randomized algorithm used for finding the minimum cut in a graph, has a time complexity of
$O(V^2)$, where V is the number of vertices in the graph.

Randomized Min-Cut Algorithm

Theorem for Min-Cut Set

The algorithm outputs a min-cut set with probability at least

$$\frac{2}{n(n-1)}.$$

Randomized Min-Cut Algorithm

Python Simulation

- Open Campuswire post regarding today's lecture (L10)
- Download the .ipynb file pasted in the comments

Thank You!
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